

Modul Internet of Things: Menggunakan STM32 + ESP01
Praktikum Senior Project, Lab Dasar 2
Departemen Teknik Elektro dan Teknologi Informasi, Fakultas Teknik,
Universitas Gadjah Mada

Pengaturan IOC pada STM32CubeIDE

1. Buat New Project sesuai dengan tipe STM32 kalian, contoh: STM32F401CDU6, sampai selesai dan tertampil file IOC kalian.
2. Klik “System Core” > RCC > High Speed Clock (HSE) set sebagai “Crystal/Ceramic Resonator”
3. Klik “System Core” > SYS > Debug set sebagai “Serial Wire”
4. Klik “Connectivity” > USART1 > Mode set sebagai “Asynchronous” dan pastikan pada Parameter Settings > Basic Parameters > Baud Rate > 115200 Bits/s
5. Klik save sampai ter-generate main.c, kemudian edit main.c berikut:

Percobaan 1: STM32+ESP01 berlaku sebagai server yang membuat its own wifi connection, dan HP/laptop terkoneksi ke wifi stm32 dengan memasukkan SSID dan Password, kemudian membuka website 192.168.51.1 dan mengendalikan LED STM32 menggunakan tombol di website tersebut.

Di user code begin 2:

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```
uint8_t rxBuffer[512] = {0};
uint8_t ATisOK;
int channel;
int onoff;
int led = 1;
char ATcommand[64];
char ATcommandB[1024];
char ATcommandN[100];
char ATcommandF[100];
char ATcommandT[16];
sprintf(ATcommandB, "<!DOCTYPE html><html>\n<head>\n\
<title>STM32 - ESP8266</title>\n<link href=\"data:image/x-icon;base64,\
A\" rel=\"icon\" type=\"image/x-icon\"><style>\nhtml {\
display: inline-block; margin: 0px auto; text-align: center;}\n\
body{margin-top: 50px;}\n.button {display: block;\n\
width: 70px;\nbackground-color: #008000;\nborder: none;\ncolor: white;\n\
padding: 14px 28px;\ntext-decoration: none;\nfont-size: 24px;\n\
margin: 0px auto 36px; \nborder-radius: 5px;}\n\
.button-on {background-color: #008000;}\n.button-on:active\
{background-color: #008000;}\n.button-off {background-color: #808080;}\n\
.button-off:active {background-color: #808080;}\n\
p {font-size: 14px;color: #808080;margin-bottom: 20px;}\n\
</style>\n</head>\n<body>\n<h1>STM32 - ESP8266</h1>");
sprintf(ATcommandN, "<p>Light is currently on\
</p><a class=\"button button-off\" href=\"/lightoff\">OFF</a>");
sprintf(ATcommandF, "<p>Light is currently off\
</p><a class=\"button button-on\" href=\"/lighton\">ON</a>");
sprintf(ATcommandT, "</body></html>");
int countB = strlen(ATcommandB);
int countN = strlen(ATcommandN);
int countF = strlen(ATcommandF);
int countT = strlen(ATcommandT);

sprintf(ATcommand, "AT+RST\r\n");
memset(rxBuffer, 0, sizeof(rxBuffer));
```

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```
HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
HAL_Delay(500);

ATisOK = 0;
while(!ATisOK){
    sprintf(ATcommand, "AT+CWMODE_CUR=2\r\n");
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 1000);
    if(strstr((char *)rxBuffer, "OK")){
        ATisOK = 1;
    }
    HAL_Delay(500);
}

ATisOK = 0;
while(!ATisOK){
    sprintf(ATcommand, "AT+CWSAP_CUR=\"STM32\", \"12345678\", 1, 3, 4, 0\r\n");
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 1000);
    if(strstr((char *)rxBuffer, "OK")){
        ATisOK = 1;
    }
    HAL_Delay(500);
}

ATisOK = 0;
while(!ATisOK){
    sprintf(ATcommand, "AT+CIPAP_CUR=\"192.168.51.1\"\r\n");
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 1000);
    if(strstr((char *)rxBuffer, "OK")){
        ATisOK = 1;
    }
    HAL_Delay(500);
}
```

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```
ATisOK = 0;
while(!ATisOK){
    sprintf(ATcommand, "AT+CIPMUX=1\r\n");
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 1000);
    if(strstr((char *)rxBuffer, "OK")){
        ATisOK = 1;
    }
    HAL_Delay(500);
}

ATisOK = 0;
while(!ATisOK){
    sprintf(ATcommand, "AT+CIPSERVER=1,80\r\n");
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 1000);
    if(strstr((char *)rxBuffer, "OK")){
        ATisOK = 1;
    }
    HAL_Delay(500);
}
```

Kemudian di dalam while(1)

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```
memset(rxBuffer,0,sizeof(rxBuffer));
HAL_UART_Receive (&huart1, rxBuffer, 512, 1000);
if(strstr((char *)rxBuffer,"+IPD,0")) channel = 0;
else if(strstr((char *)rxBuffer,"+IPD,1")) channel = 1;
else if(strstr((char *)rxBuffer,"+IPD,2")) channel = 2;
else if(strstr((char *)rxBuffer,"+IPD,3")) channel = 3;
else if(strstr((char *)rxBuffer,"+IPD,4")) channel = 4;
else if(strstr((char *)rxBuffer,"+IPD,5")) channel = 5;
else if(strstr((char *)rxBuffer,"+IPD,6")) channel = 6;
else if(strstr((char *)rxBuffer,"+IPD,7")) channel = 7;
else channel = 100;

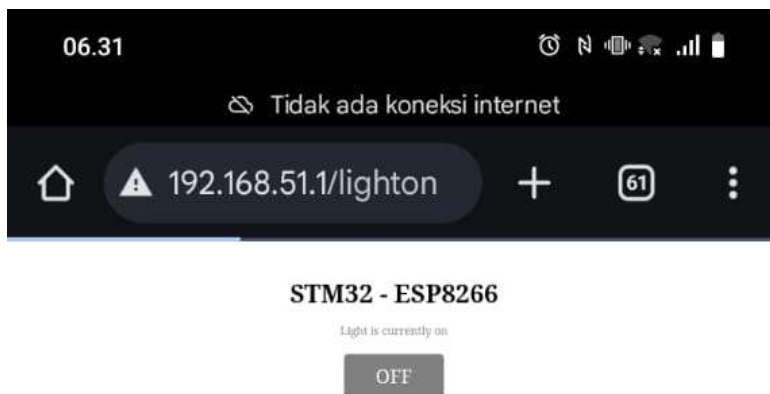
if(strstr((char *)rxBuffer,"GET /lighton")) onoff = 0;
else if(strstr((char *)rxBuffer,"GET /lightoff")) onoff = 1;
else onoff = led;

if(channel<8 && onoff == 1)
{
    HAL_GPIO_WritePin(GPIOC, GPIO_PIN_13, GPIO_PIN_SET);
    led = 1;
    sprintf(ATcommand,"AT+CIPSEND=%d,%d\r\n",channel,countB+countF+countT);
    memset(rxBuffer,0,sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1,(uint8_t *)ATcommand,strlen(ATcommand),1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
    if(strstr((char *)rxBuffer,">"))
    {
        memset(rxBuffer,0,sizeof(rxBuffer));
        HAL_UART_Transmit(&huart1,(uint8_t *)ATcommandB,countB,1000);
        HAL_UART_Transmit(&huart1,(uint8_t *)ATcommandF,countF,1000);
        HAL_UART_Transmit(&huart1,(uint8_t *)ATcommandT,countT,1000);
        HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
    }
    sprintf(ATcommand,"AT+CIPCLOSE=%d\r\n",channel);
```


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```
memset(rxBuffer, 0, sizeof(rxBuffer));
HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
channel=100;
}
else if(channel<8 && onoff == 0)
{
    HAL_GPIO_WritePin(GPIOC, GPIO_PIN_13, GPIO_PIN_RESET);
    led = 0;
    sprintf(ATcommand, "AT+CIPSEND=%d,%d\r\n", channel, countB+countN+countT);
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
    if(strstr((char *)rxBuffer, ">"))
    {
        memset(rxBuffer, 0, sizeof(rxBuffer));
        HAL_UART_Transmit(&huart1, (uint8_t *)ATcommandB, countB, 1000);
        HAL_UART_Transmit(&huart1, (uint8_t *)ATcommandN, countN, 1000);
        HAL_UART_Transmit(&huart1, (uint8_t *)ATcommandT, countT, 1000);
        HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
    }
    sprintf(ATcommand, "AT+CIPCLOSE=%d\r\n", channel);
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
    channel=100;
}
```

Koneksikan wifi hp/laptop Anda ke wifi dengan SSID: STM32 dan password 12345678 dan buka browser dan search link 192.168.51.1 sehingga akan tampil halaman web berikut:



Klik button ON agar LED di STM32 menyala, dan klik OFF agar LED di STM32 padam.

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Percobaan 2: STM32+ESP01 berlaku sebagai client terkoneksi ke Tethering Internet HP, kemudian publish data ke broker test.mosquitto.org, sedangkan Komputer/laptop berlaku sebagai client 2 dan subscribe data dari broker, dengan topic yang sama dengan yang si-publish oleh STM32+ESP01.

```
23 /* USER CODE BEGIN Includes */
24 #include "string.h"
25 #include "stdio.h"
26 /* USER CODE END Includes */
```

Kemudian di dalam int main(void){

```
81  /* USER CODE BEGIN Init */
82  void send_AT_command(const char *cmd, uint32_t delay_ms) {
83      HAL_UART_Transmit(&huart1, (uint8_t *)cmd, strlen(cmd), HAL_MAX_DELAY);
84      HAL_Delay(delay_ms);
85  }
86
87  void send_and_receive(const char *cmd, uint32_t delay_ms) {
88      uint8_t buffer[512] = {0};
89
90      HAL_UART_Transmit(&huart1, (uint8_t *)cmd, strlen(cmd), HAL_MAX_DELAY);
91      HAL_Delay(delay_ms);
92
93      HAL_UART_Receive(&huart1, buffer, sizeof(buffer), 1000);
94      printf("%s\n", buffer); // Display received response on serial terminal
95  }
96
97  void ESP01_ConnectWiFi(const char *ssid, const char *password) {
98      send_and_receive("AT+RST\r\n", 3000); // Reset ESP
99      HAL_Delay(2000); // Wait for boot
100
101      send_and_receive("AT+CWMODE=1\r\n", 1000); // Set mode: Station (client)
102
103      char wifiCommand[128];
104      snprintf(wifiCommand, sizeof(wifiCommand),
105              "AT+CWJAP=\"%s\", \"%s\"\r\n", ssid, password);
106
107      send_and_receive(wifiCommand, 8000); // Connect to Wi-Fi network
108  }
```

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```
109 void send_AT(const char *cmd, uint32_t delay_ms) {
110     HAL_UART_Transmit(&huart1, (uint8_t *)cmd, strlen(cmd), HAL_MAX_DELAY);
111     HAL_Delay(delay_ms);
112 }
113
114 void send_data(uint8_t *data, uint16_t len) {
115     HAL_UART_Transmit(&huart1, data, len, HAL_MAX_DELAY);
116     HAL_Delay(500); // small delay after sending data
117 }
118
119 void ESP01_ConnectMQTTBroker(const char *broker, uint16_t port) {
120     char cmd[128];
121     snprintf(cmd, sizeof(cmd),
122              "AT+CIPSTART=\"TCP\", \"%s\", %d\r\n", broker, port);
123     send_AT(cmd, 5000);
124 }
125
126 void ESP01_Send_MQTT_CONNECT(const char *clientID) {
127     uint8_t packet[128];
128     uint8_t clientID_len = strlen(clientID);
129     uint8_t total_len = 10 + 2 + clientID_len;
130
131     uint8_t index = 0;
132     packet[index++] = 0x10; // CONNECT command
133     packet[index++] = total_len; // Remaining length
134     packet[index++] = 0x00; packet[index++] = 0x04; // Length of "MQTT"
135     packet[index++] = 'M'; packet[index++] = 'Q'; packet[index++] = 'T'; packet[index++] = 'T';
136     packet[index++] = 0x04; // Protocol level 4
137     packet[index++] = 0x02; // Clean session
138     packet[index++] = 0x00; packet[index++] = 0x3C; // Keep alive = 60 sec
139     packet[index++] = 0x00; packet[index++] = clientID_len;
140     memcpy(&packet[index], clientID, clientID_len);
141     index += clientID_len;
142
143     // Send CIPSEND first
144     char sendCmd[32];
145     snprintf(sendCmd, sizeof(sendCmd),
146              "AT+CIPSEND=%d\r\n", index);
147     send_AT(sendCmd, 1000);
148 }
```


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```
148
149     // Send packet data
150     send_data(packet, index);
151 }
152
153 void ESP01_Send_MQTT_PUBLISH(const char *topic, const char *message) {
154     uint8_t packet[256];
155     uint8_t topic_len = strlen(topic);
156     uint8_t message_len = strlen(message);
157
158     uint8_t remaining_len = 2 + topic_len + message_len;
159     uint8_t index = 0;
160
161     packet[index++] = 0x30;           // PUBLISH command
162     packet[index++] = remaining_len; // Remaining length
163     packet[index++] = 0x00;
164     packet[index++] = topic_len;
165     memcpy(&packet[index], topic, topic_len);
166     index += topic_len;
167     memcpy(&packet[index], message, message_len);
168     index += message_len;
169
170     // Send CIPSEND first
171     char sendCmd[32];
172     sprintf(sendCmd, sizeof(sendCmd),
173             "AT+CIPSEND=%d\r\n", index);
174     send_AT(sendCmd, 1000);
175
176     // Send packet data
177     send_data(packet, index);
178 }
179 /* USER CODE END Init */
180
181
182 /* USER CODE BEGIN 2 */
183 HAL_Delay(2000);
184 ESP01_ConnectWiFi("vivo Y28", "oke123456");
185 ESP01_ConnectMQTTBroker("test.mosquitto.org", 1883);
186
187 HAL_Delay(2000);
188
189 ESP01_Send_MQTT_CONNECT("STM32Client");
190 HAL_Delay(2000);
191
192 /* USER CODE END 2 */
193
194 /* USER CODE BEGIN WHILE */
195 while (1)
196 {
197     ESP01_Send_MQTT_PUBLISH("/sudiyo", "temperature=30\r\n");
198 }
199 /* USER CODE END WHILE */
```

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Buka Command Prompt: Windows + R ketik cmd ketik mosquito_sub -h test.mosquitto.org -t "/topic"
--

Percobaan 3: